

ALOK KUMAR

AI-ML Engineer

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Summary

Computer Science undergraduate (CGPA 9.29) specializing in applied AI/ML, with hands-on experience building and deploying computer vision and NLP systems — including a real-time object-detection pipeline (YOLOv8) and an NLP-based resume-matching engine — alongside full-stack MERN development.

Skills

AI/ML & Data Science: Machine Learning, Computer Vision (YOLOv8, OpenCV, Pose Estimation), NLP, Data Analysis, Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn, Gradio, Streamlit

Programming & Querying: Python, Java, SQL

Web & Full-Stack: MongoDB, Express.js, React, Node.js, HTML, CSS, Bootstrap, JavaScript

Developer Tools: Git, GitHub, VS Code

Core CS Foundations: OOPs, DBMS, Operating Systems, Computer Networks

DSA (Java): Array, String, Linked List, Stack, Queue, HashMap

Projects

Aqua Wings: Drowning Detection System | *Python, OpenCV, YOLOv8* 🔗 [Live Demo](#) May 2026

- Engineered a real-time drowning detection system using YOLOv8 and pose estimation for distress detection.
- Developed anomaly detection to identify abnormal body movements and distinguish drowning from normal swimming.
- Trained and validated the model on Roboflow-annotated video datasets with high precision and low false positives.
- Integrated real-time alert system and evaluated performance for deployment readiness on surveillance feeds.

AI Resume Analyzer & Job Matcher | *Python, Flask, NLP* 🔗 [Live Demo](#) Feb 2026

- Built a full-stack Resume Analyzer using Flask and SQLAlchemy with dashboards for candidates and recruiters.
- Implemented NLP-based skill extraction and resume-job matching using a weighted scoring algorithm.
- Designed a relational database to store applications, match scores, and skill-gap analysis.
- Developed a secure file upload pipeline with validation and automated resume processing.

WanderLust – Property Booking Web App (MERN) | *MongoDB, Express.js, Node.js* 🔗 [Live Demo](#) Oct 2025

- Built a MERN-stack Airbnb-like web app with property listings, category filters, and GST-based INR pricing.
- Implemented secure authentication with protected routes, session management, and user-based listings.
- Designed a responsive React UI with real-time search and multi-category browsing.
- Developed a Node.js + Express backend (MVC) with MongoDB and deployed on Render.com.

Internship

Data Science & AI/ML Intern | 🔗 [Live Demo](#) June 2025 – July 2025

YBI Foundation

Tools Used: Python, Scikit-learn, Pandas, NumPy

- Developed a Bank Customer Churn Prediction model to identify high-risk customers.
- Performed data preprocessing, feature engineering, SMOTE balancing, and feature scaling.
- Identified key churn factors through feature importance analysis.
- Optimized model performance using GridSearchCV and evaluation metrics such as F1-score and ROC-AUC.

Education

B.Tech in Computer Science and Engineering 2023 – 2027

Amity University Haryana

CGPA: 9.29

Senior Secondary (Class XII) – CBSE 2021 – 2022

SCR Public School

Percentage: 88.66%

Secondary (Class X) – CBSE 2019 – 2020

SCR Public School

Percentage: 91.16%

Awards & Certifications

- AI Program Conducted by Samsung Innovation Campus | 🔗 [View Certificate](#)
- Generative AI Foundations using Google Cloud | 🔗 [View Certificate](#)
- Data Science and AI/ML Internship Certificate | 🔗 [View Certificate](#)
- Delta – Full Stack Web Development Certification | 🔗 [View Certificate](#)